



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO PALKARAR K. PONNU AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

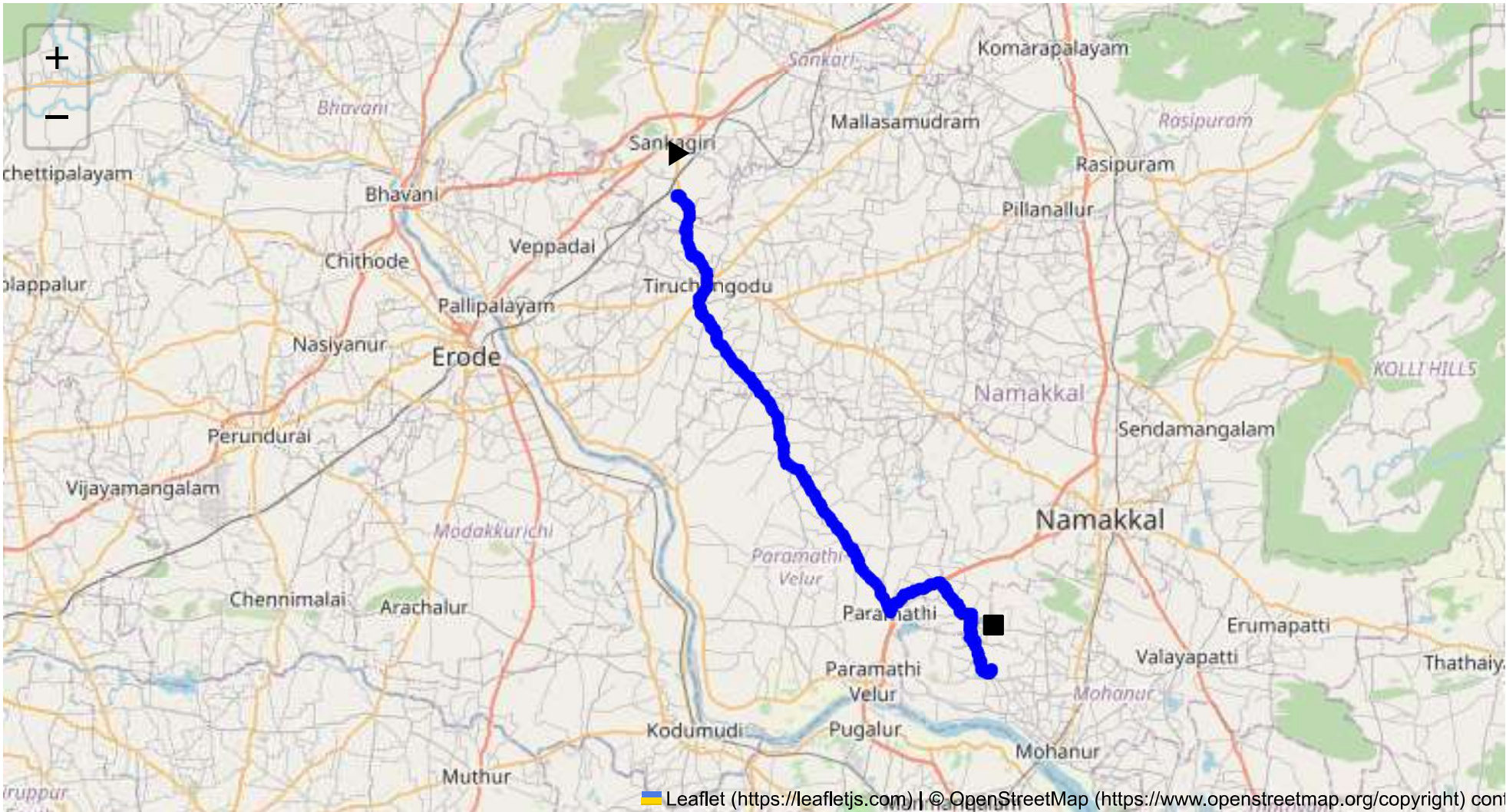
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 51.93 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.1164187, 78.09187539)



Welcome to the Journey Risk Management Study

Route Analysis from CVQF+23W, Sangagiri to 438R+HQJ, Vallipuram - Palapatti Road

- 1. Overview of the Route Map:

- The route spans approximately 51.93 kilometers generally moving southeast through rural and semi-urban areas in Tamil Nadu. The journey is expected to take around 1.20 hours under normal conditions, with major segments likely involving state highways and local roads.

2. Typical Weather Conditions and Hazards:

- The region experiences a tropical climate. The monsoon season (June to September) can bring heavy rains, leading to potential flooding and road washouts. Heat waves are common in the summer (April to June), which can affect vehicle performance. Fog may be an intermittent issue during winter months.

3. Traffic Patterns:

- Peak traffic is typically observed in the morning (7-10 AM) and evening (5-8 PM). Congestion may occur near market areas and junctions, particularly in small towns and villages along the route.

4. Road Quality and Infrastructure:

- State highways are generally well-maintained but narrow in sections. Potholes and uneven surfaces may be encountered, especially on secondary and local roads. Be cautious of livestock and pedestrians potentially straying onto these roads.

5. Alternative Routes for Emergencies:

- In case of road blockages or accidents, consider using smaller connecting roads through nearby villages. However, these are typically less maintained and should be navigated carefully.

6. Local Regulations on Hazardous Material Transport:

- Adhere to Tamil Nadu's regulations for transporting hazardous materials including necessary permits. Transport vehicles must be properly labeled, and routes for hazardous materials transport may have restrictions in populated zones.

7. Historical Incidents:

- Past incidents often involve overturned vehicles due to narrow roads and improper turning maneuvers. Areas with a high frequency of accidents should be marked as caution zones.

8. Environmental Considerations:

- Sensitive areas include agricultural lands and occasional protected forest zones near the route. Ensure minimal environmental disruption by avoiding spills and disposing of waste properly.

9. Communication Coverage:

- Major sections of the route should have adequate mobile coverage due to proximity to urban centers. However, rural stretches may experience signal dead zones; satellite communication devices are recommended in these areas.

10. Estimated Emergency Response Times:

- Emergency response may vary; within urban or semi-urban centers, expect a response within 30-45 minutes. Remote areas might experience delays of 1 hour or more due to distance from emergency facilities.

11. Overall Summary of Risk Assessment:

- The route involves moderate risk, primarily from weather conditions, road quality, and interactions with high-density traffic zones. Planning for these contingencies, including pre-identified detours and maintaining updated communication tools, will mitigate negative impacts. Training drivers for specific hazards along with adherence to local regulations will ensure safer transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Turn	High	11.37868, 77.89498	15 KM/Hr
5	Turn	High	11.37850, 77.89453	15 KM/Hr
6	Turn	Medium	11.16764, 78.01565	30 KM/Hr
7	Blind Spot	Blind Spot	11.15573, 78.02176	10 KM/Hr
8	Turn	Medium	11.16243, 78.02715	30 KM/Hr
9	Turn	High	11.16241, 78.02729	15 KM/Hr
10	Turn	High	11.17605, 78.05538	15 KM/Hr
11	Turn	Medium	11.17447, 78.05732	30 KM/Hr
12	Turn	Medium	11.17225, 78.05906	30 KM/Hr
13	Turn	Medium	11.16071, 78.06874	30 KM/Hr
14	Turn	Medium	11.15756, 78.07014	30 KM/Hr
15	Turn	Medium	11.15582, 78.07004	30 KM/Hr
16	Turn	Medium	11.15501, 78.07151	30 KM/Hr
17	Turn	Medium	11.15437, 78.07194	30 KM/Hr
18	Blind Spot	Blind Spot	11.15411, 78.07842	10 KM/Hr
19	Turn	High	11.14790, 78.07733	15 KM/Hr
20	Turn	Medium	11.14736, 78.07811	30 KM/Hr
21	Turn	Medium	11.14715, 78.07821	30 KM/Hr
22	Turn	Medium	11.14327, 78.07733	30 KM/Hr
23	Turn	Medium	11.13737, 78.07789	30 KM/Hr
24	Turn	Medium	11.13608, 78.07979	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
25	Blind Spot	Blind Spot	11.13449, 78.08021	10 KM/Hr
26	Turn	High	11.13464, 78.08069	15 KM/Hr
27	Turn	High	11.11376, 78.09124	15 KM/Hr
28	Blind Spot	Blind Spot	11.11657, 78.09219	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	college	PGP Group of Educational Institutions	11.1748266, 78.0472484	30 km/h	Medium
18	college	PGP College of Engineering	11.1748402, 78.0466235	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.37850, 77.89453



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16764, 78.01565



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16243, 78.02715



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.16241, 78.02729



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.17605, 78.05538



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.17447, 78.05732



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.17225, 78.05906



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.16071, 78.06874



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15756, 78.07014



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15582, 78.07004



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15501, 78.07151



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.15437, 78.07194



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.15411, 78.07842



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14790, 78.07733



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.14736, 78.07811



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.14715, 78.07821



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.14327, 78.07733



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.13737, 78.07789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.13608, 78.07979



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.13449, 78.08021



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.13464, 78.08069



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.11376, 78.09124



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.11657, 78.09219
